SEARCHING FOR VARIABILITY IN SPECTROPHOTOMETRIC STANDARD STARS

1 Scientific Justification and Perceived Impact

An accurate measure of stellar brightness in physical units is essential for a multitude of scientific endeavors by most astronomical telescopes. This is accomplished through observations of spectrophotometric standard stars. The HST/CALSPEC[1] database of flux standards contains many stars with Hubble spectral coverage in the 0.115–1 μm range with some extensions to longer wavelengths of 1.7 or 2.5 μm. These CALSPEC standards also have flux distributions modeled out to 32 μm to enable JWST flux calibration. They also enable cross-instrument and cross telescope calibration, tying data from *JWST* to *HST*, *Spitzer*, and other ground-based telescopes [2].

The importance of absolute flux calibration with a precision of ≈1% should not be underestimated, as this precision is crucial for some of the most important discoveries in astronomy. For example, interpretations of the supernova data that measure the accelerating expansion rate of the universe depend critically on uncertainties in flux calibration [12, 13, 11]. JWST will exploit new methods of determining the distances to supernova-hosting galaxies in the near-infrared in order to better understand the Hubble Tension, such as short-period oxygen-rich Miras [3, 4] and dust-free carbon stars [8, 6], These methods will require absolute flux calibration tied with precision to HST. Precise calibrations are also necessary to accurately measure the size and temperature of exoplanet-hosting stars [14], which in turn is necessary to understand the demographics of exoplanetary orbits and atmospheres.

The HST/CALSPEC database is recognized as a preferred source of standard stars for absolute flux calibration of a wide variety of space and ground-based instrumentation [2]. Many observatory calibrations rely on this database, and indeed, the JWST calibration stars were selected from this list. However, the flux standards are only viable if the stars are constant in brightness, and *TESS* is the premier mission for vetting these stars at previously unexplored time scales and precision.

TESS is able to find low-amplitude variability that has been previously difficult to measure for these stars, though is important to identify for the highest precision calibrations. Even though the CALSPEC stars have been scrutinized for stability using available observations, a recent paper [10] used TESS to show that 15 of 37 of the candidate JWST calibration stars were variable, with 4 showing variability at levels greater than 0.5 percent, preventing them from acting as good flux standards for high-precision photometry (see Figure 1). We request TESS data at a 2-minute cadence of CALSPEC stars that have yet to be observed by TESS. At the same time we request continued monitoring of the JWST subset of CALSPEC stars that showed no variability in order to ensure that there is no change in their behavior as JWST begins observations and executes its first year of photometric calibrations.

While most of the anticipated large-amplitude variability we expect to find occurs at periods longer than 10 minutes, we request 2-minute cadence for two reasons. First, the 2-minute data products enable us to easily and accurately measure the amplitude of the variability. Specifically, the PDC light curves from the 2-minute data correct the amplitude

[1] http://www.stsci.edu/hst/instrumentation/reference-data-for-calibration-and-tools/astronomical-catalogs/calspec

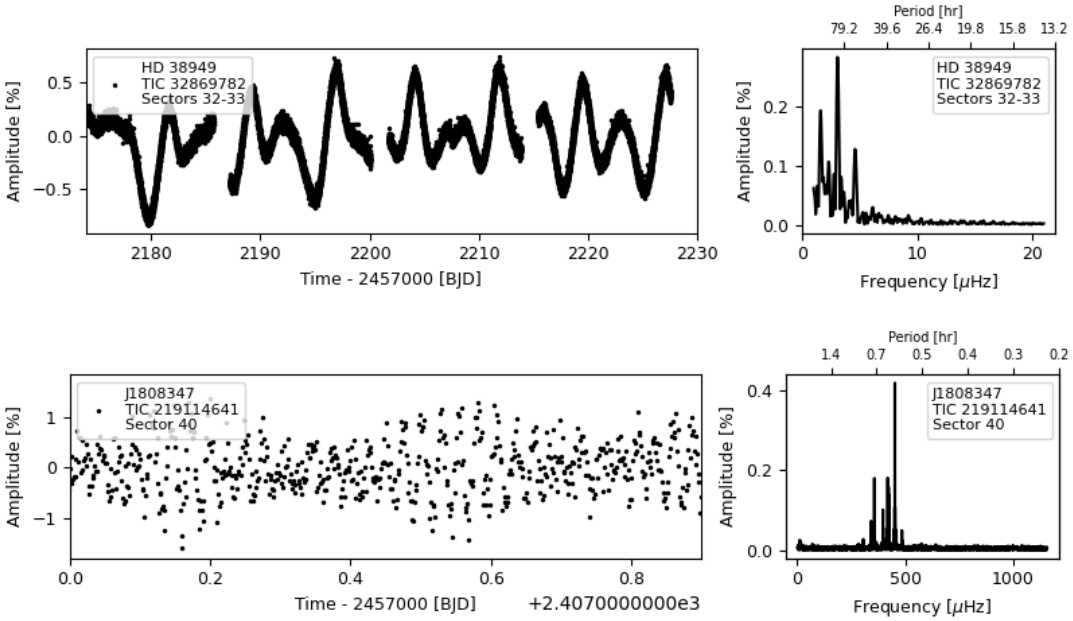


Figure 1: *TESS* light curve and periodogram of the *JWST* candidate calibration stars, HD 38949 (TIC 32869782) and TIC 219114641. These stars were removed from the JWST calibration standard lists based on these observations [10].

of the variations for crowding and apply a robust co-trending algorithm. Second, certain stellar variability can occur at short periods, including delta Scuti pulsations, stellar rotation and flares [9, 1, 5]. Indeed, TESS has already revealed short-period variability in some of the CALSPEC standard stars [10].

2 Analysis Plan and Technical Feasibility

We provide a list of 62 CALSPEC standard stars that have not been known to vary at large amplitudes. Given the current TESS cycle 5 fields, 20 are observable during Cycle 5, 4 of which have never been observed by TESS before. We provide the entire list in case the TESS pointings change, with priority given to the JWST calibration stars that have yet to be observed by TESS. The targets range in *TESS* magnitudes from 4 to 17, with a median of 11 mag. We plan to analyze the variability of these targets using the Lightkurve software package [7]. We will look for both periodic variability and sudden changes in brightness using periodograms, transit search algorithms, and visual inspection of the light curves. The amplitudes and characteristic periods of any measured variability will be published and publicly noted on the HST/CALSPEC webpage hosted at the Mikulski Archive for Space Telescopes.

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3 Three Example Targets

TIC ID	Common Name	RA (deg)	Dec (deg)	TESS mag	Obj. Type	Comments
426959594	2M0036+18	9.06705	18.3528	15.92	L3.5	2-min
441630081	BD+75 325	124.8440	54.08600	9.87	O5p	2-min
452740376	BD+28 4211	327.7959	28.863	10.87	sdO	2-min